18.704 Expository Paper

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This is an expository paper produced for 18.704, a CI-M at MIT. This loosely follows Chapter 9 of Serre's *An Introduction to Finite Groups*. This paper expects basic knowledge of groups, rings, and fields, at approximately the level of an introductory undergraduate course. Some elementary number theoretic results are also used without proof.

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1 Introduction

The purpose of this paper is to explore finite subgroups of $\mathrm{GL}_n(F)$ where F is some field. Here, we take $\mathrm{GL}_n(F)$ to be the set of invertible $n \times n$ matrices over F. It can also be seen as the set of invertible (or equivalently injective, or equivalently surjective) maps from F^n to F^n.

One natural question that ultimately has a rather trivial answer is the following, which provides motivation for later work.

Question 1. Given a group G of order m, is there some n such that $G \leq \mathrm{GL}_n(F)$?

It turns out that this question has a simple answer. We know by Cayley's theorem that $G \leq S_m$, and the subgroup of $m \times m$ permutation matrices (over any field F) is isomorphic to S_m. Thus, we can take $n = m$.

The goal of this paper, consequently, is to explore what groups G occur for small n, and to attempt to control their size and structure.

Notation. For the remainder of this paper, p and ℓ will denote distinct primes and F a field. We will take $0 \in \mathbb{N}$ al a Bourbaki. We will consider $\mathrm{GL}_n(F)$ and assume $n \geq 1$. In general, we write $g, h \in \mathrm{GL}_n(F)$ for an element of a subgroup we are considering and write $A, B, M, U, V \in \mathrm{GL}_n(F)$ when thought of primarily as a matrix.

1.1 Recollection of necessary results

Proposition 2 (Cayley-Hamilton)**.** Any matrix is a root of its characteristic polynomial.

Proposition 3 (Dirichlet)**.** Suppose $\gcd(a, m) = 1$. There are infinitely many primes congruent to a modulo m.

Proposition 4. The (monic) minimal polynomial of any (resp. every) primitive nth root of unity ζ_n over $\mathbb{C}$ is

$$\Phi_n(x) \;=\; \prod_{m \in (\mathbb{Z}/n\mathbb{Z})^\times} (x - \zeta_n^m) \;\in\; \mathbb{Z}[x]. \tag{1}$$

Section 2 will cover finite subgroups of $\mathrm{GL}_n(\mathbb{Q})$, working towards a theorem of Minkowski. Then, Section 3 will cover a natural generalization of this to $\mathrm{GL}_n(\mathbb{C})$ which was originally proved by Jordan.

2 Finite subgroups of $\mathrm{GL}_n(\mathbb{Q})$

This section will describe finite subgroups of $\mathrm{GL}_n(\mathbb{Q})$. Obviously, enumerating all possible finite subgroups requires enumerating all finite groups. Similarly, counting the number of such subgroups possible requires enumerating the number of groups of a certain order. The goal, therefore, will be to bound the size of each individual subgroup as a function of n.

Observation 5. Let $g \in \mathrm{GL}_n(\mathbb{Q})$ have exact order k. Let $f(x) \in \mathbb{Q}[x]$ be the (monic) minimal polynomial of g. We know that $f(x) \mid x^k - 1$ (so f has distinct roots) but for all $m < k$ we have $f(x) \nmid x^m - 1$. Now, since $f(x)$ is fixed by all elements of $\mathrm{Aut}(\bar{\mathbb{Q}}/\mathbb{Q})$, we know that if $f(x)$ has a

primitive mth root of unity as a root it has all the primitive mth roots of unity as roots. Thus, $f(x)$ must have as a root (over $\mathbb{C}$) some primitive c_ith root of unity for $\operatorname{lcm}(c_i) = k$, and we also need $\sum \varphi(c_i) \leq n$.

Now, this is useful for the $n = 2$ case, because the only such n that are possible are $n \in \{1, 2, 3, 4, 6\}$ (why?). Additionally, for p prime we have $\varphi(p) = p - 1$, so we get the following corollary.

Corollary 6. The only prime numbers that can divide the order of a finite subgroup must be those that are less than or equal to $n + 1$.

2.1 Exploration of $\mathrm{GL}_2(\mathbb{Q})$

The $n = 1$ case is trivial and left as an exercise to the reader. The next simplest case, therefore, is the $n = 2$ case. Let $G \leq \mathrm{GL}_2(\mathbb{Q})$ be a finite subgroup.

Observation 7. We have $|G| = 2^s 3^t$ for some $s, t \in \mathbb{N}$ by Corollary 6.

Let $H \leq G$ be a Sylow 3-subgroup. Motivated that $\mathbb{Z}/9\mathbb{Z} \not\leq H$ we explore what groups H can be. It is easy to see that $H \cong \{1\}$ and $H \cong \mathbb{Z}/3\mathbb{Z}$ are both possible:

Example 8. Consider $g_3 := \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. It is easy to check that $g^3 = 1$. Note that this is the matrix associated to $\zeta_3 \in \mathbb{Q}(\zeta_3)$ with the standard basis $\{1, \zeta_3\}$.

Our next goal is to show that $(\mathbb{Z}/3\mathbb{Z})^{\oplus 2} \leq H$ cannot occur. One way to show this is the following two results.

Exercise 9. Let $g \in G$ have exact order 3. Then, there is some invertible matrix M such that MGM^{-1} maps g to g_3.

Proposition 10. Let $h \in \mathrm{GL}_2(\mathbb{Q})$ be exact order 3 and $g_3 h = h g_3$. Then $h = g_3$ or $h = g_3^2$.

Proof. Let $h = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ be such a matrix. Observe that its minimal polynomial divides $x^3 - 1$ and $h \neq 1$, so divides $x^2 + x + 1$ (which is irreducible over $\mathbb{Q}$). Then

$$g_3 h = \begin{bmatrix} -c & -d \\ a - c & b - d \end{bmatrix} \quad \text{and} \quad h g_3 = \begin{bmatrix} b & -a - b \\ d & -c - d \end{bmatrix} \tag{2}$$

If we set these equal, we conclude that $b = -c$ and $d = a - c = a + b$. Thus, we have $h = \left(\begin{smallmatrix} a & b \\ -b & a+b \end{smallmatrix}\right)$. Now, we have $-1 = \operatorname{tr}(h) = 2a + b$ and $1 = \det(h) = a^2 + ab + b^2$. Solving this nets $a = 0, b = -1$ or $a = -1, b = 1$, which give $h = g_3$ and $h = g_3^2$ respectively. $\square$

Corollary 11. For $G \leq \mathrm{GL}_2(\mathbb{Q})$, the Sylow 3-subgroup H of G is either trivial or isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

Proof. If the Sylow 3-subgroup has order > 3 then it has a subgroup of order 9. The only groups of order 9 are $\mathbb{Z}/9\mathbb{Z}$ and $(\mathbb{Z}/3\mathbb{Z})^{\oplus 2}$, neither of which can be a subgroup of $\mathrm{GL}_2(\mathbb{Q})$. $\square$

We now explore the possibilities for s. We now let $K \leq G$ be the Sylow 2-subgroup, and $A \leq K$ be a maximal abelian normal subgroup.

Lemma 12. An element $g \in \mathrm{GL}_2(F)$ has order ≤ 2 if and only if $g = \pm I_2$ or $\det g = -1$ and $\operatorname{tr} g = 0$.

Proof. Let $g^2 = 1$. Then the minimal polynomial of g divides $x^2 - 1$. Thus, either $g = \pm I_2$, or the minimal polynomial is exactly $x^2 - 1$. If so, the characteristic polynomial is also $x^2 - 1$, so $\mathrm{tr} g = 0, \det g = -1$. □

Exercise 13. Suppose $\sqrt{-1} \notin F$. Then $g \in \mathrm{GL}_2(F)$ has exact order 4 if and only if $\det g = 1$ and $\mathrm{tr} g = 0$. In particular, $g^2 = -I_2$.

Corollary 14. We have $(\mathbb{Z}/2\mathbb{Z})^{\oplus 3} \not\leq \mathrm{GL}_2(F)$.

Proof. Observe that elements of exact order 2 have determinant -1 or are $-I_2$. Suppose that there are distinct elements g_1, g_2, g_3 of exact order 2 whose pairwise products are exact order 2. If none of these are $-I_2$ then $\det g_1 g_2 = 1 = \det g_1 g_3$, so $g_1 g_2 = -I_2 = g_1 g_3$, and thus $g_2 = g_3$. This is a contradiction.

If $(\mathbb{Z}/2\mathbb{Z})^{\oplus 3} \leq \mathrm{GL}_2(\mathbb{Q})$ then there are at least 6 elements which are not $\pm I_2$ and picking any such three contradicts the previous paragraph. □

Remark 15. The same proof shows that if $(\mathbb{Z}/2\mathbb{Z})^{\oplus 2} \leq G$ then this subgroup is $\{\pm I_2, \pm g\}$ for some $g \neq -I_2$ of exact order 2. Consequently, if $4 \mid \#G$ then $-I_2 \in G$.

Thus, A must be isomorphic to one of

$$\{1\},\ \mathbb{Z}/2\mathbb{Z},\ \mathbb{Z}/4\mathbb{Z},\ (\mathbb{Z}/2\mathbb{Z})^{\oplus 2},\ \text{or } \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}. \tag{3}$$

Once again it is easy to find examples of the first four. We work to eliminate the last group. By similar logic to Lemma 12, any element of order 4 has minimal polynomial $x^2 + 1$, and thus has determinant 1. Thus, if we let g_1 be a element of order 2 other than $-I_2$ and g_2 be an element of order 4. Then $g_1 g_2$ is also order 4, and

$$1 \;=\; \det g_1 g_2 \;=\; \det g_1 \det g_2 \;=\; (-1)(1) \;=\; -1. \tag{4}$$

Proposition 16. Let G be a nilpotent group and $A \trianglelefteq G$ be a maximal abelian normal subgroup. Then $A = C_G(A)$ and G/A is isomorphic to a subgroup of $\mathrm{Aut}(A)$.

Proof. The first equality is implied by supersolvability, which nilpotency implies. Consider $\varphi : G \to \mathrm{Aut}(A)$ via $g \mapsto (a \mapsto gag^{-1})$, and the kernel is $C_G(A) = A$, and by the first isomorphism theorem we are done. □

Thus $K/A \cong K/C_K(A)$ is isomorphic to a subgroup of $\mathrm{Aut}(A)$. By the following table, we conclude that $|K| \leq 8$, so $|G| \leq 24$. See the next subsection for the complete classification of what G can be.

Group A	$\mathrm{Aut}(A)$	$\#\mathrm{Aut}(A)$
$\{1\}$	$\{1\}$	1
$\mathbb{Z}/2\mathbb{Z}$	$\{1\}$	1
$\mathbb{Z}/4\mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z}$	2
$(\mathbb{Z}/2\mathbb{Z})^{\oplus 2}$	S_3	6
D_8	D_8	8

Table 1: Automorphism groups of relevant groups.

To summarize, we have proved the following theorem which we generalize to higher n in §2.3.

Theorem 17. Let $G \leq \mathrm{GL}_2(\mathbb{Q})$ be a group of finite order. Then $\#G \mid 24$.

2.2 Complete classification of finite subgroups of $\mathrm{GL}_2(\mathbb{Q})$

This subsection will fully enumerate all possible finite subgroups $G \leq \mathrm{GL}_2(\mathbb{Q})$. Specifically, write $D_{2n} = \langle r, s | r^n = s^2 = rsrs = 1 \rangle$. Then G is one of

$$\{1\},\ \mathbb{Z}/2\mathbb{Z},\ \mathbb{Z}/3\mathbb{Z},\ \mathbb{Z}/4\mathbb{Z},\ \mathbb{Z}/6\mathbb{Z},\ (\mathbb{Z}/2\mathbb{Z})^{\oplus 2},\ D_6,\ D_8,\ \text{or } D_{12}. \tag{5}$$

Examples in the abelian case are obvious. For examples for the non-abelian case, think of these matrices as (conjugate to) rotations of $\mathbb{R}^2$ and the matrix (conjugate to) $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ will be a reflection. It remains to show no other groups are possible, which Propositions 18 and 19 together imply.

Proposition 18. Suppose $1 + r^2 + s^2 = 0$ does not have a solution in F. Then the quaternion group Q_8 is not a subgroup of $\mathrm{GL}_2(F)$. In particular, $Q_8 \not\leq \mathrm{GL}_2(\mathbb{Q})$.

Proof. Observe that if $\sqrt{-1} \in F$ then $(r, s) = (\sqrt{-1}, 0)$ has $1 + r^2 + s^2 = 0$. Thus, by Remark 15 we have $-I_2 \in G$. This is in the center of G. Thus, we need matrices A, B such that

$$A^2 \ = \ B^2 \ = \ (AB)^2 \ = \ -I_2. \tag{6}$$

Suppose

$$A \ = \ \begin{bmatrix} a & b \\ \frac{-1-a^2}{b} & -a \end{bmatrix}, \tag{7}$$

and consider conjugating G by

$$P \ = \ \begin{bmatrix} 0 & b \\ 1 & -a \end{bmatrix} \quad \text{so that} \quad P^{-1}AP \ = \ \begin{bmatrix} a/b & 1 \\ 1/b & 0 \end{bmatrix} \begin{bmatrix} b & 0 \\ -a & -1 \end{bmatrix} \ = \ \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \ =: \ M. \tag{8}$$

Thus, by working with PGP^{-1}, we may assume without loss of generality G contains $\pm I_2, \pm M$. Now, put

$$B \ = \ \begin{bmatrix} r & s \\ \frac{-1-r^2}{s} & -r \end{bmatrix} \quad \text{and compute} \quad MB \ = \ \begin{bmatrix} \frac{1+r^2}{s} & r \\ r & s \end{bmatrix}, \tag{9}$$

and we need MB to be exact order 4 and thus trace 0. Thus, $\frac{1+r^2}{s} + s = 0$ so $1 + r^2 + s^2 = 0$. $\square$

Proposition 19. Suppose $\#G \in \{12, 24\}$. Then $G \cong D_{12}$.

The proof of this proposition amounts to the remainder of the subsection. Ease of comprehension of the proof will benefit from familiarity with nilpotent subgroups. We first recall the following lemma:

Lemma 20. Let G be a finite group. The following are equivalent

(a) G is nilpotent

(b) Every Sylow p-subgroup in G is normal

(c) G is a direct product of its Sylow p-subgroups

Consequently, any p-group will be nilpotent.

We now introduce the notion of the Fitting subgroup and will then analyze the structure of $F(G)$.

Definition 21. For a finite group G, the *Fitting subgroup* $F(G)$ is the unique maximal normal nilpotent subgroup of G.

It exists because the product of two normal nilpotent subgroups must be normal and nilpotent:

Theorem 22 (Fitting)**.** If M and N are nilpotent normal subgroups of a group G, then their product MN is also a nilpotent normal subgroup of G.

Proof. Normality is well-known. To show nilpotency, we show that every Sylow p-subgroup is normal. Fix some p and let R be a Sylow p-subgroup of $M \cap N$. Now, R is contained in the unique Sylow p-subgroup $P \subseteq M$ (resp. $Q \subseteq N$). Now, P is characteristic in M (it is the only subgroup of its size) and thus normal in G. Likewise, $Q \trianglelefteq G$. Thus $PQ \leq MN$ is a normal subgroup of G of size

$$\frac{\#P\#Q}{\#(P \cap Q)} = \frac{\#P\#Q}{\#R}, \tag{10}$$

and this is the exact size we are looking for. Thus, $AB \trianglelefteq MN$ is a Sylow p-subgroup, and thus MN is nilpotent. □

We need one final lemma which will help restrict the structure of the Fitting subgroup.

Lemma 23. Let $x \in G$ be exact order 3 or 4. Then $C_G(x)$ is cyclic.

Proof. The minimal polynomial $m_x(t)$ of x is either $t^2 + t + 1$ or $t^2 + 1$. Then $\mathbb{Q}[x]$ as a sub-algebra of $M_2(\mathbb{Q})$ is isomorphic to $\mathbb{Q}[t]/m_x(t)$. Note that the torsion elements of $\mathbb{Q}[x]$ are therefore either μ_4 or μ_6, and thus torsion is cyclic. Thus, it suffices to show that the elements in $M_2(\mathbb{Q})$ that commute with x are those in $\mathbb{Q}[x]$.

To finish, let $M_2(\mathbb{Q})$ act on $V = \mathbb{Q}^2$, and consider how $\mathbb{Q}[x]$ acts by restriction. This gives us a $\mathbb{Q}[x]$ action making V a 1-dimensional vector space over $\mathbb{Q}[x]$. Now, suppose that A acts on V and commutes with x. Then A commutes with $\mathbb{Q}[x]$ and thus lies is in $\mathrm{End}_{\mathbb{Q}[x]}(V) = \mathbb{Q}[x]$. □

Proof of Proposition 19. Let $F = F(G)$ be the Fitting subgroup of G. Remark 15 and the fact that $4 \mid \#G$ imply $-I_2 \in G$, so F contains a central (in G) element of order 2. We split into cases based off if F contains an element of order 3 or not.

First, suppose that F contains an element h of order 3. Since $F(G)$ is nilpotent, by applying Corollary 11 to F we know $h \in Z(F)$ so $C_F(h) = F$. Then Lemma 23 and Observation 5 tell us F is cyclic of order 6. Next, consider some $g \notin F$ and let it act on F by conjugation. If this action was trivial, then $F' = \langle F, g\rangle$ would be a nilpotent subgroup (g is in its center, and the quotient by g is isomorphic to some quotient of F) of size either 12 or 24. Thus F' has index ≤ 2 and is therefore a larger normal nilpotent subgroup, contradiction.

Thus, assume every $g \notin F$ acts non-trivially on F. Now, the only non-trivial automorphism of F is order 2, and thus g^2 acts trivially (so $g^2 \in F$). Thus G/F is isomorphic to $\mathbb{Z}/2\mathbb{Z}$ or $(\mathbb{Z}/2\mathbb{Z})^{\oplus 2}$. In the latter case, let $g, g' \in G$ be in different F-cosets that are not F. Then gg' must act trivially, but is not in F. Thus $G/F \cong \mathbb{Z}/2\mathbb{Z}$ which implies $G \cong D_{12}$.

Instead suppose that F does not contain an element of order 3. Let $h \in G$ be order 3, and once again let h act on F by conjugation. If h acts non-trivially, then $3 \mid \mathrm{Aut}(F)$. We know that F is a 2-group, and consulting Table 1, we get $F \cong (\mathbb{Z}/2\mathbb{Z})^{\oplus 2}$ and thus h acts transitively on non-identity

elements. Fortunately, we also know F contains $-I_2$, and this must be fixed by conjugation by h, contradiction.

Thus, assume that h acts trivially. In this case, $F' = \langle F, h\rangle$ is some nilpotent subgroup of G of order at least 6. Since $F' \supsetneq F$ we know F' is not normal and thus index at least 3. Thus, $\#G = 24$ and $\#F' = 6$, and thus $\#F = 2$. We now consider the number $n_2 \in \{1, 3\}$ of Sylow 2-subgroup(s) of G. If $n_2 = 1$ then said subgroup is a nilpotent normal subgroup so $\#F \geq 8$, contradiction. Thus, $n_2 = 3$ and define $\varphi : G \to S_3$ via how conjugation permutes the Sylow 2-subgroups. The transitive subgroups of S_3 are order 3 and 6 so Sylow's 2nd theorem tells us $\#\mathrm{Ker}(\varphi) \in \{24/3, 24/6\}$. This is a normal nilpotent subgroup, contradicting that $\#F = 2$. □

2.3 Minkowski's Theorem

We now switch our attention to larger n. Classification of all possible subgroup is now unwieldy, but we can still bound the size of possible subgroups.

Let $\lfloor x \rfloor$ denote the largest integer $\leq x$ and define

$$M(n, \ell) = \sum_{i=0}^{\infty} \left\lfloor \frac{n}{(\ell - 1)\ell^i} \right\rfloor \quad \text{and} \quad M(n) \;=\; \prod_{\ell} \ell^{M(n,\ell)}. \tag{11}$$

The goal of the rest of this section is to prove the following theorem.

Theorem 24 (Minkowski)**.** Let $G \leq \mathrm{GL}_n(\mathbb{Q})$ be finite. Then (i) $\#G \mid M(n)$ and (ii) $M(n)$ is the minimal such number with this property.

We can compute

$$\begin{aligned} M(0) \;&=\; 1,\; M(1) \;=\; 2 \cdot 1,\; M(2) \;=\; 24,\; M(3) \;=\; 2 \cdot 24,\; M(4) \;=\; 5760, \\ M(5) \;&=\; 2 \cdot 5760,\; M(6) \;=\; 2903040,\; M(7) \;=\; 2 \cdot 2903040, \end{aligned} \tag{12}$$

where $M(2) = 24$ is as we computed in the prior subsection. Observe that, for ℓ odd, $\ell - 1$ is even and thus for even n we have $M(n, \ell) = M(n + 1, \ell)$, and $M(n, 2) + 1 = M(n + 1, 2)$.

Observation 25. To show (i) it suffices by Sylow's theorem that any ℓ-subgroup of $\mathrm{GL}_n(\mathbb{Q})$ has size bounded by $\ell^{M(n,\ell)}$.

Fix ℓ and G a finite ℓ-subgroup of $\mathrm{GL}_n(\mathbb{Q})$. Since there are finitely many elements in G, we can find some m such that $G \subseteq \mathrm{GL}_n(\mathbb{Z}[1/m])$ (e.g. matrices M where Mm^k and $M^{-1}m^k$ have integer coefficients for some $k \in \mathbb{N}$). Now, for $p \nmid m$ we get a natural map $\mathrm{GL}_n(\mathbb{Z}[1/m]) \to \mathrm{GL}_n(\mathbb{F}_p)$ by reducing each coefficient modulo p.

Lemma 26. For p sufficiently large, the composition

$$G \to \mathrm{GL}_n(\mathbb{Z}[1/m]) \to \mathrm{GL}_n(\mathbb{F}_p) \tag{13}$$

is injective.

Proof. Put d as the least common multiple of all the denominators. For each $M \in G \setminus \{I_n\}$ pick some coefficient c_M of the matrix $d(M - I_n)$ which is nonzero (and integral). Now, when $p \nmid dc_M$ the kernel of $G \to \mathrm{GL}_n(\mathbb{F}_p)$ does not contain M. Thus, for $p > |d \prod_{M \neq 1} c_M|$, we are done. □

Corollary 27. For the same p as in the previous lemma, $\#A$ divides $r_n(p) = \prod_{i=1}^{n}(p^i - 1)$.

The order of $\mathrm{GL}_n(\mathbb{F}_p)$ is $r_n(p)p^{n(n-1)/2}$. Our goal is now to find some p that minimizes the ℓ-adic valuation of $r_n(p)$. The following results completes (i) for ℓ odd.

Lemma 28. Let ℓ be an odd prime and $x \in \mathbb{Z}$ such that the equivalence class of x modulo ℓ^2 generates $(\mathbb{Z}/\ell^2\mathbb{Z})^\times$. Then

$$\nu_\ell\left(\prod_{i=1}^{n}(x^i - 1)\right) \;=\; M(n,\ell), \tag{14}$$

where ν_ℓ is the ℓ-adic valuation (i.e. $\nu_\ell(x)$ is the largest positive integer k such that $\ell^k \mid x$).

Proof. Since x is a generator of $(\mathbb{Z}/\ell\mathbb{Z})^\times$, if $\ell - 1$ does not divide i then $x^i \not\equiv 1 \pmod{\ell}$ and otherwise put $i = (\ell-1)m$ so that $y := x^{\ell-1} = 1 + \ell z$ for some $z \in \mathbb{Z}$ with $\ell \nmid z$. Then put $m = r\ell^a$. We then have

$$x^i \;\equiv\; y^m \;\equiv\; (y^{\ell^a})^r \;\equiv\; (1+\ell^{a+1}z)^r \;\equiv\; 1 + \ell^{a+1}zr \pmod{\ell^{a+2}}, \tag{15}$$

and thus $\nu_\ell(x^i - 1) = a + 1 = \nu_\ell(m) + 1$. To finish, observe that

$$\begin{aligned}\nu_\ell\left(\prod_{i=1}^{n}(x^i - 1)\right) \;=\; \sum_{i=1}^{n}\nu_\ell(x^i - 1) \;&=\; \sum_{m=1}^{s}\nu_\ell(x^m - 1)\\ &=\; \sum_{m=1}^{s}\nu_\ell(m) + 1 \;=\; m + \nu_\ell(s!) \;=\; M(n,\ell),\end{aligned} \tag{16}$$

where $s = \lfloor m/(\ell-1)\rfloor$ and we use $\lfloor\lfloor a/b\rfloor/c\rfloor = \lfloor a/(bc)\rfloor$ for $b, c \in \mathbb{Z}^+$. □

By Proposition 3, we can find a prime p congruent to x modulo ℓ^2. Thus, to finish (i), we only need to consider the Sylow 2-subgroup.

2.4 Local Analysis ($\ell = 2$)

In this case, the embedding into $\mathrm{GL}_n(\mathbb{F}_p)$ is not sufficient because it has too many factors of 2. Instead, we show that we can embed G into the orthogonal group $O_n(\mathbb{F}_p)$ for all sufficiently large p. Observe that to do so, it suffices to show that we can embed G into some conjugate (in $\mathrm{GL}_n(\mathbb{F}_p)$) of $O_n(\mathbb{F}_p)$. We use the standard averaging argument, which is used extensively, for example in the proof of Maschke's theorem.

Theorem 29. Let p be sufficiently large. Then G embeds into some conjugate of $O_n(\mathbb{F}_p)$. That is, there is some matrix $Q \in \mathrm{GL}_n(\mathbb{F}_p)$ with

$$G \xrightarrow{\text{reduce modulo } p} QO_n(\mathbb{F}_p)Q^{-1} \xrightarrow{M \mapsto Q^{-1}MQ} O_n(\mathbb{F}_p) \tag{17}$$

injective.

Proof. From Lemma 26 we embed into $\mathrm{GL}_n(\mathbb{F}_p)$. So it suffices to show G maps into $QO_n(\mathbb{F}_p)Q^{-1}$ for some fixed Q.

Let $V = \mathbb{Q}^n$ be the vector space and $\langle \cdot, \cdot \rangle_0$ denote its standard inner product defined by $\langle u, v \rangle_0 = u^T v$. Motivated by Maschke's Theorem, consider

$$\langle u, v \rangle \;=\; \frac{1}{|G|} \sum_{g \in G} \langle gu, gv \rangle_0 \;=\; \frac{1}{|G|} \sum_{g \in G} u^T g^T g v. \tag{18}$$

Now for

$$M \;=\; \frac{1}{|G|} \sum_{g \in G} g^T g, \tag{19}$$

we have $\langle u, v \rangle = u^T M v$. It is now easy to show that M is positive definite and thus $\langle \cdot, \cdot \rangle$ is a G-invariant inner product.

If we pick p such that M is also invertible modulo p then $\langle \cdot, \cdot \rangle$ descends to an inner product over $\mathbb{F}_p^n$. Therefore G maps into some conjugate of the orthogonal group by picking a basis such that $\langle \cdot, \cdot \rangle$ is the standard inner product. □

By page 141 of The Geometry of the Classical Groups, we have for $n = 2m + 1$ odd

$$\#O_n(\mathbb{F}_p) \;=\; 2p^{m^2} \prod_{i=1}^{m} \left(p^{2i} - 1\right). \tag{20}$$

We now consider $p \equiv \pm 3 \pmod 8$ so that $p^2 \equiv 9 \pmod{16}$. Then, for $k = r2^a$ for r odd, we have

$$p^{2k} \;\equiv\; (p^2)^{r2^a} \;\equiv\; (1 + 2^{3+a})^r \;\equiv\; 1 + r2^{3+a} \;\equiv\; 1 + 2^{3+a} \pmod{2^{4+a}}. \tag{21}$$

Thus, the 2-adic valuation when $p \equiv \pm 3 \pmod 8$ is

$$1 + \sum_{i=1}^{m} 3 + \nu_2(i) \;=\; 1 + 3m + \nu_2(m!) \;=\; n + m + \sum_{i=1}^{\infty} \lfloor m/2^i \rfloor \;=\; M(n, 2). \tag{22}$$

To finish, when n is even, observe that if G is a subgroup of $\mathrm{GL}_n(\mathbb{Q})$ then we can embed $\mathrm{GL}_n(\mathbb{Q}) \hookrightarrow \mathrm{GL}_{n+1}(\mathbb{Q})$, and thus G is a subgroup of $\mathrm{GL}_{n+1}(\mathbb{Q})$. Then $2\#G \mid M(n+1) = 2M(n)$ because $\{\pm I_{n+1}\} \times G \leq \mathrm{GL}_{n+1}(\mathbb{Q})$. Thus, the result for $n + 1$ implies the result for n.

Remark 30. By working with $O_n(\mathbb{F}_p)$ we effectively eliminate the ℓ-adic terms coming from $p^{\mathrm{odd}} - 1$. This also works for $\ell > 2$, but does not help since we can (and do!) pick p such that this is non-zero modulo ℓ.

2.5 Proof of (ii)

It suffices to find a $G \leq \mathrm{GL}_n(\mathbb{Q})$ with $\nu_\ell(\#G) = M(n, \ell)$. Let $m = \lfloor n/(\ell - 1) \rfloor$ so that $M(n, \ell) = M(m(\ell - 1), \ell)$. Thus, it suffices to show the result for $m(\ell - 1)$. Let $\mathbb{Z}/\ell\mathbb{Z}$ act faithfully on $\mathbb{Q}^\ell$ by rotating the coordinates. Elements of the form $(t, t, \ldots, t)$ are fixed, and let V_0 be the quotient by this subspace. Thus, we get a faithful action

$$\mathbb{Z}/\ell\mathbb{Z} \to \mathrm{GL}(V_0). \tag{23}$$

Now take $V = \oplus_{i=1}^m V_0$ (i.e. the direct sum of m copies of V_0) where $(\mathbb{Z}/\ell\mathbb{Z})^{\oplus m}$ acts coordinate-wise on each summand and S_m acts by permuting the m summands. Then $H = (\mathbb{Z}/\ell\mathbb{Z})^{\oplus m} \rtimes S_m$ acts faithfully on V, and $\dim V = m(\ell - 1)$. Put $G = \mathrm{im} H \leq \mathrm{GL}(V)$, and we have

$$\nu_\ell(\#G) \;=\; \nu_\ell(\ell^m m!) \;=\; m + \nu_\ell(m!) \;=\; M(m(\ell - 1), \ell). \tag{24}$$

3 Finite subgroups of $\mathrm{GL}_n(\mathbb{C})$

Of course, there are roots of unity of arbitrary order for $n = 1$. We therefore cannot hope for any statement equivalent to Minkowski's theorem. Instead, however, one notices that G often contains a large abelian subgroup, motivating the following theorem.

Theorem 31 (Jordan)**.** There exists a function $f(n)$ such that every finite subgroup of $\mathrm{GL}_n(\mathbb{C})$ contains an abelian normal subgroup of index bounded by $f(n)$.

We will show that $f(n) = (2\sqrt{2n} + 1)^{2n^2}$ suffices, although much better bounds are known. For example, one can take $f(2) = 60$ while our bound is much larger: $5^8 = 390625$. Specifically, one can show that

$$\mathrm{SL}_2(\mathbb{F}_5) \;\leq\; \mathrm{SU}(2) \;\leq\; \mathrm{GL}_2(\mathbb{C}), \tag{25}$$

and $A_5 \cong \mathrm{PSL}_2(\mathbb{F}_5) := \mathrm{SL}_2(\mathbb{F}_5)/\{\pm I_2\}$ so that $f(2) \geq 60$. Showing $f(2) \leq 60$ is much more difficult.

To prove Theorem 31, we will first explore Hermitian spaces, and then find an abelian normal subgroup by analyzing the unitary group $U_n(\mathbb{C})$. The ultimate bound will be given by a packing argument.

3.1 Hermitian spaces

For the remainder of this section, V will denote an n-dimensional vector space with a positive definite Hermitian form written $\langle \cdot, \cdot \rangle$. This induces a norm $|x| = \langle x, x \rangle^{1/2}$. For $A \in M_n(\mathbb{C})$ there is a unique matrix A^*, called the adjoint, such that

$$\langle Ax, y \rangle \;=\; \langle x, A^* y \rangle \tag{26}$$

for all $x, y \in V$. If we choose an orthonormal basis, then $A^* = \bar{A}^T$. We call an A *normal* if $AA^* = A^*A$. If $A = A^*$, we say that A is *Hermitian*, which of course implies normal. If further $\langle Ax, x \rangle \geq 0$ for all x, we call A *positive Hermitian*. We recall the following linear-algebraic result.

Proposition 32. Let A be normal. There is an orthonormal basis of V made up of eigenvectors of A.

Finally, recall that a matrix is *unitary* if $AA^* = 1$, or equivalently, A preserves the inner product. Consequently, these matrices make a subgroup $U_V \subseteq \mathrm{GL}(V)$, and under the standard Hermitian form, we have $U_V = U_n(\mathbb{C})$.

Definition 33. Let $A \in M_n(\mathbb{C})$. Define the *norm* $\mathcal{N}(A)$ of A to be

$$\mathrm{tr}(AA^*) \;=\; \mathrm{tr}(A^*A) \;=\; \sum_{ij} |a_{ij}|^2 \;=\; \sum_i |Ae_i|^2. \tag{27}$$

The following proposition now outlines some properties of the norm.

Proposition 34. Let A be normal, $B \in M_n(\mathbb{C})$ and U, V unitary. We have

(a) $\mathcal{N}(1) = n$

(b) $\mathcal{N}(AU) = \mathcal{N}(UA) = \mathcal{N}(A)$

(c) Suppose A has eigenvalues λ_i then $\mathcal{N}(A) = \sum |\lambda_i|^2$

(d) $\mathcal{N}(AB - BA) \leq 2\mathcal{N}(A)\mathcal{N}(B)$

(e) $\mathcal{N}(I_n - UVU^{-1}V^{-1}) \leq 2\mathcal{N}(1-U)\mathcal{N}(1-V)$

(f) Put $V_0 = V$ and $V_m = UV_{m-1}U^{-1}V_{m-1}^{-1}$. Then $\mathcal{N}(I - V_m) \leq 2^m\mathcal{N}(1-U)^m\mathcal{N}(1-V)$.

Proof. (a), (b), and (c) are trivial and (f) follows by induction and (e). (e) follows from (d) by

$$\mathcal{N}(I - UVU^{-1}V^{-1}) \;=\; \mathcal{N}(VU - UV) \;=\; \mathcal{N}((I-V)(I-U) - (I-U)(I-V)) \tag{28}$$

and observing that normal matrices are closed under addition. Thus, it remains to show (d).

Pick an orthonormal basis with eigenvalues λ_i. Then if $B = (b_{ij})$ we have $AB - BA = ((\lambda_i - \lambda_j)b_{ij})$. Thus

$$\mathcal{N}(AB - BA) \;=\; \sum_{ij} |\lambda_i - \lambda_j|^2 |b_{ij}|^2, \tag{29}$$

and thus it suffices to observe that

$$|\lambda_i - \lambda_j|^2 \;\leq\; |\lambda_i - \lambda_j|^2 + |\lambda_i + \lambda_j|^2 \;=\; 2(|\lambda_i|^2 + |\lambda_j|^2) \;\leq\; 2\mathcal{N}(A).$$

□

Now, our normal abelian subgroup is ensured by the following proposition.

Proposition 35. Let U be the set of all $U \in U_V$ such that $\mathcal{N}(1-U) < 1/2$ and $G \leq U_V$ be finite and put H as the subgroup generated by $G \cap U$. Then H is a normal abelian subgroup of G.

Proof. Normality is insured by $\mathcal{N}(V(1-U)V^{-1}) = \mathcal{N}(1-U) < 1/2$, while commutativity of elements in $G \cap U$ is insured by the following proposition. □

Proposition 36. Let U, V be elements of G such that $\mathcal{N}(1-U), \mathcal{N}(1-V) < 1/2$. Then U and V commute.

Proof. Put $V_0 = V$ and $V_m = UV_{m-1}U^{-1}V_{m-1}^{-1}$. Then $\mathcal{N}(I - V_m) \leq 2^m\mathcal{N}(1-U)^m\mathcal{N}(1-V) \to 0$ by Proposition 34. Thus, since V_m can only take on finitely many values, eventually $V_m = I$, so U and V_m commute for some m. By inducting downwards on m, the following lemma completes the proof. □

Lemma 37. Let U, V be unitary with $\mathcal{N}(I - V) < 2$. If $UVU^{-1}V^{-1}$ commutes with U then U and V commute.

Proof. Let $M = VUV^{-1}$, and observe that U and M commute (U commutes with M^{-1}) and are conjugate. By Proposition 32 there is an orthonormal basis $\{e_1, \ldots, e_n\}$ which are eigenvectors of both M and U. Let the eigenvalues for M be λ_i and for U be μ_i. If $\lambda_i = \mu_i$ for all i then $M = U$ so U and V commute. Otherwise, let i such that $\lambda_i \neq \mu_i$. The vector Ve_i is now an eigenvector of M with eigenvalue $\lambda_i \neq \mu_i$. Thus, Ve_i and e_i are orthogonal, so

$$|(I - V)e_i|^2 \;=\; |e_i - Ve_i|^2 \;=\; |e_i|^2 + |Ve_i|^2 \;=\; 2, \tag{30}$$

so $\mathcal{N}(I - V) \geq 2$, contradiction. □

3.2 Bounding the number of spread out points on a sphere

From the prior subsection, we know that $H \trianglelefteq G$ is abelian. Let $S \subseteq G$ be a set of representatives of elements of G/H. If $x \neq y$ are in S then $x^{-1}y \notin H$ so $\mathcal{N}(1 - x^{-1}y) \geq 1/2$ so $\mathcal{N}(x - y) \geq 1/2$.

Now, we can embed $\mathrm{GL}_n(\mathbb{C})$ into $\mathbb{R}^{2n^2}$ coordinatewise (and $x + iy \mapsto (x, y)$). This turns $\mathcal{N}$ into the standard Euclidean norm. Thus, points in S are distance $\geq (2n)^{-1/2}$ apart. The following proposition now shows that

$$[G : H] \;=\; \#S \;\leq\; (2\sqrt{2n} + 1)^{2n^2}. \tag{31}$$

Proposition 38. Let $0 < \epsilon < 1$ and $X \in S^{d-1} \subseteq \mathbb{R}^d$ be such that points of X are $\geq 2\epsilon$ apart. Then

$$|X| \;\leq\; \left(\epsilon^{-1} + 1\right)^d. \tag{32}$$

Proof. Define $B_y(r)$ be the set of points within distance r of y. Then the $B_x(\epsilon)$ for $x \in X$ are disjoint and open, and are contained in the ball $B_0(1 + \epsilon)$. Since volume scales independently of center point and with the dth power of radius, we are done. $\square$

This completes the proof of Theorem 31. This gives us control on the index of maximal abelian normal subgroups of $\mathrm{GL}_n(\mathbb{C})$. It provides a nice analog to Theorem 24, which controls the size of subgroups of $\mathrm{GL}_n(\mathbb{Q})$.